

Probabilistic Life Expectancy Projections for Puerto Rico Using the Lee-Carter Model

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Introduccion

Background

Mortality projections serve as a foundational pillar in population forecasting, standing alongside fertility and migration as one of the three critical variables that determine the shape of our future society(Preston, 2000). They are essential for understanding future demographic trends and planning for social services, healthcare, and economic development(Kontis et al., 2017). They are a critically important tool for policymakers, demographers, and researchers to anticipate and respond to changes in population dynamics. The failure to accurately project mortality rates can lead to significant misallocations of resources and inadequate planning for future needs.

Within the balancing equation of demography, mortality serves as the primary metric for assessing the health and survival limits of a population. As global societies undergo a the epidemiological transition characterized by a shift from infectious diseases to chronic and age-related illness, the ability to forecast future survival patterns has become essential for understanding broader demographic

trends (Omram, 2001; Vaupel, 2010). Consequently, mortality projections are no longer merely statistical exercises; they are vital instruments for social planning, economic forecasting, and the development of public health strategies.

For policymakers and researchers, these projections provide the quantitative foundation required to anticipate and respond to the challenges of an aging society. Specifically, accurate forecasts are critical for the solvency of national pension systems and the longterm sustainability of social security frameworks (Bengtsson & Keilman, 2019; R. D. Lee & Carter, 1992). In the area of public health, projections allow for the strategic allocation of resources toward geriatric care and the management of non-communicable diseases that are common in older populations (Christensen et al., 2009). Without robust data, governments risk “longevity risk” the economic threat posed by populations living significantly longer than anticipated by historical models (Cairns et al., 2006).

The historical record demonstrates that failing to accurately project mortality can lead to severe structural consequences. For decades, official forecasts consistently underestimated the pace of mortality decline, particularly at older ages, leading to inadequate planning for healthcare infrastructure and retirement funding (Booth, 2006; Oeppen & Vaupel, 2002). This historical underestimation underscores the necessity of moving beyond simple deterministic models toward modern computerized tools and stochastic frameworks.

Advances in Mortality Modeling

The transition from deterministic to stochastic modeling represents a paradigm shift in the field of demographic forecasting. Historically, mortality projections relied on deterministic methods, such as the Gompertz-Makeham law of mortality or expert led “scenario based” variants used by institutions like the United States Social Security Administration. These older models often assumed a fixed rate of improvement or relied on subjective “high,” “medium,” and “low” scenarios that failed to capture the non-linear and age-specific nature of mortality decline (Booth, 2006). Critically, deterministic models offer no formal measure of uncertainty; they provide a single trajectory that historically has proven to be consistently over-conservative, leading to the repeated underestimation of life expectancy gains at older ages (R. Lee & Miller, 2001).

The landmark introduction of the Lee-Carter model revolutionized the field by shifting the focus from expert opinion to a data-driven, statistical framework. The Lee-Carter method employs a stochastic approach that decomposes the logarithm of age-specific death rates into three components: an average age profile, a time varying mortality index, and an age-specific response to that index. By using Singular Value Decomposition (SVD) to extract the primary trend of mortality change, the model allows the data itself to dictate the pace of decline rather than imposing arbitrary limits on survival (R. D. Lee & Carter, 1992). This method has since become the “gold standard” in demography (Giroso & King, 2008).

Beyond its predictive accuracy, the primary value of the Lee-Carter frame-

work lies in its stochastic nature. Unlike deterministic models, which offer a “best guess,” stochastic models treat future mortality as a random process with an associated probability distribution. By modeling the time-varying index (k_t) as a stochastic time series typically a random walk with drift—researchers can generate probabilistic intervals for future life expectancy. This quantification of uncertainty is indispensable for modern risk management; it allows policymakers and actuaries to calculate “longevity risk” and the probability of extreme outcomes, such as a “95% confidence interval” for the future old-age dependency ratio (Cairns et al., 2006; R. Lee, 2000). In a 21st-century context, where the pace of medical innovation and the impact of global health crises are unpredictable, the ability to communicate uncertainty through these probabilistic frameworks is no longer a luxury, but a prerequisite for responsible fiscal and social planning.

Puerto Rico: Demographic Context and Gap in the Literature

Policymakers and public health officials in Puerto Rico need annual stochastic mortality projections as part of producing annual probabilistic population projections of the island population. The lack of precise, forward looking mortality data has already manifested in severe institutional challenges across Puerto Rico, most notably within its social safety nets and fiscal frameworks. The island’s retirement systems offer a cautionary example of how demographic uncertainty can lead to

systemic collapse. For decades, the mismatch between projected and actual dependency ratios, driven by a shrinking tax base and a rapidly aging population, contributed to the total exhaustion of the employees retirement system's trust funds (Abel & Deitz, 2014; Cox Alomar, 2019). Because policymakers lacked robust, stochastic projections to account for the accelerated pace of aging, the government was eventually forced to transition to a "Pay As You Go" or "PayGo" model. This shift has placed an enormous, immediate burden on the central budget, where pension payments now compete directly with essential services like education and public safety (Waldman, 2023). Furthermore, the inability to accurately project mortality trends has complicated the island's long term recovery and debt restructuring efforts. Strategic documents, such as the certified fiscal plans issued by the Financial Oversight and Management Board, rely heavily on population forecasts to determine Puerto Rico's future debt paying capacity (Puerto Rico Fiscal Agency and Financial Advisory Authority 2024)

This thesis will use the Lee-Carter model to first analyze the series from 1950 to 2023 into its three components, the a_x , b_x , and k_t parameters by age and sex to obtain a deeper understanding of the underlying mortality trends in Puerto Rico. Then, we will bootstrap the parameters to create a distribution of possible future mortality scenarios, which will allow us to create a stochastic forecast of mortality rates and life expectancy at birth for the next 40 years. Finally, we will compare our projections to those produced by the UN World Population Prospects, the U.S. Census Bureau's International Data Base, and the University of Washington's Institute for Health Metrics and Evaluation to contextualize our findings within the

broader landscape of demographic forecasting and assess the robustness of the model, identify sources of uncertainty and bias in the projection and provide evidence based recommendations for improving future Puerto Rico specific mortality forecasting efforts

Literary Review

Lee Carter

The Lee-carter model was introduced in 1992 by Ronald Lee and Lawrence Carter in their paper “Modeling and Forecasting U.S. Mortality”(R. D. Lee & Carter, 1992). It revolutionized mortality forecasting by providing a data-driven approach in lieu of the expert-led methods that were previously dominant. Prior to its development, official forecasts often underestimated the gains in human life expectancy (Olshansky, 1988), creating a pressing need for more accurate projections to assess the solvency of pension systems and social security programs.

The model can be described simply with the equation $\ln(m_{x,t}) = a_x + b_x k_t + e_{x,t}$. More verbose, it states that the natural log of the mortality rates of a given year $\ln(m_{x,t})$ are equivalent to the baseline mortality a_x plus the product of the age specific sensitivity to changes in mortality b_x and the time varying mortality index k_t plus an error term $e_{x,t}$. To achieve this, Lee and Carter applied Singular Value Decomposition (SVD) to extract the characteristic pattern of mortality improvement over time. The solution to the SVD is not unique, so they imposed

two constraints: the sum of b_x across ages must equal 1, and the sum of K_t across time must equal 0. This allowed them to identify a unique solution that captured the dominant trend in mortality improvement. Then by only taking the first singular value and its corresponding vectors in a_x and b_x , they reduce the complexity of the data while retaining the most important information. This serves as compression of the data reducing the number of parameters needed to model mortality trends but keeping the essential features of the data. Following this, the time-varying mortality index K_t is modeled as a random walk with drift, allowing for the creation of objective prediction intervals that provided a quantifiable measure of uncertainty. This was a significant departure from previous “best-guess” forecasts, which lacked any formal way to assess the range of possible outcomes.

The simplicity and robustness of the Lee-Carter model allowed it to outperform contemporary models(Booth, 2006), closely tracking actual mortality data while older models significantly under-predicted life expectancy gains. Its ability to capture the linear trend of log-mortality decline across the 20th century made it particularly effective for long-term forecasting. Unlike earlier parametric models that focused on the biological shape of the mortality curve at a single point in time, the Lee-Carter model prioritized the dynamic movement of that curve over decades.

Thanks to the nature of the Lee-Carter model extensions and variations have been developed to address its limitations and improve its accuracy. For example, the LC-ER (Extended Rotation) method was developed to account for “rotation” in mortality decline, which addresses implausible age-pattern artifacts in vanilla LC projections. Additionally, the stmomo R package(Villegas et al., 2018), whose

implementation we will be using later, provides a flexible framework for mortality forecasting using the Lee-Carter model as well as several other models, allowing users to fit the model to various data types and generate forecasts while assessing uncertainty.

The model and its extensions have been used in various applications, including forecasting breast cancer mortality in China and Pakistan (Mubarik et al., 2023), where it outperformed simpler models for the screen-age/late-onset population (ages 50–84), revealing rising mortality in Pakistan versus declining rates in China, which was used to inform targeted healthcare interventions. The UN employs the LC model to forecast global mortality trends, especially for long-term projections up to 2100 (Wpp, 2022). For high-life-expectancy countries like Japan, the model was extended to account for “rotation” in mortality decline (slowing improvements at young ages and accelerating declines at older ages). This adjustment, termed the LC-ER (Extended Rotation) method, addressed implausible age-pattern artifacts in vanilla LC projections.

Now, when talking about the implementation we will be using, Stochastic Mortality Modeling (stMoMo) is an R package that provides a comprehensive framework for fitting and forecasting mortality models, including the Lee-Carter model. Traditionally, mortality projections followed deterministic frameworks — producing “high,” “medium,” and “low” scenario variants that, while intuitive, offer no coherent probabilistic interpretation and whose resulting ranges for vital rates and population measures are inherently inconsistent with one another (R. D. Lee & Tuljapurkar, 1994). Recognizing these limitations, the field has progressively shifted toward

stochastic approaches, in which uncertainty is explicitly modeled and forecasts are expressed as probability distributions over future outcomes. This transition has been formalized at the institutional level: the 2015 edition of the UN World Population Prospects marked a landmark departure from scenario-based projections toward fully probabilistic ones, reflecting a broader consensus that only stochastic methods can fully and coherently quantify forecasting uncertainty (Alexander & Raftery, 2024) The stMoMo package allows users to easily fit the Lee-Carter model to their mortality data, generate forecasts, and assess uncertainty through confidence intervals. It also includes various extensions of the Lee-Carter model, such as the LC-ER method, which can be used to improve the accuracy of forecasts for high-life-expectancy countries. The package is designed to be user-friendly and flexible, making it accessible for researchers and practitioners in the field of mortality forecasting. For our use case, we will be using the base Lee-Carter model and using its initial fit to bootstrap its parameters to then simulate numerous forecasts to create a distribution of possible outcomes. For this we need to set several parameter. First would be the link function, which determines the random component associated with the mortality model. The default link function is the log link, this assumes that the deaths follow a Poisson distribution, which is appropriate for count data like mortality. Secondly, we need to set the restraints for the model to ensure identifiability. For this we will set it as “sum”. This ensures that $\sum_t k_t = 0$. With these two we can now create the seed fit for the bootstrapping process. The bootstrap itself requires its own set of parameters, these being the number of bootstrap samples, the approach to be applied and the type of death to resample.

Methodology

The input matrix

Starting off with the input matrix, this is the matrix of log age specific death rates that we will be using to fit the model. The matrix is organized by age and year, with the log of the death rates for each age group and year. Our original input data for this matrix comes in the form of death counts and population exposures, which we then convert into death rates by dividing the death counts by the population exposures. Finally, we take the log of these death rates to create the input matrix for the Lee-Carter model. The data comes from multiple sources. The population estimates come from the US Census Bureau decennial censuses divided into age groups and sex and interpolated for the intercensal years using a linear spline. The mortality data is a collage of multiple sources (Puerto Rico Department Of Health, WHO database and the Center for Disease Control and Prevention. The data is organized into 5-year age groups and single year of age for the first year of life, and then 85+ for the last group.

We then pass this matrix to the stMoMo package to fit the Lee-Carter model

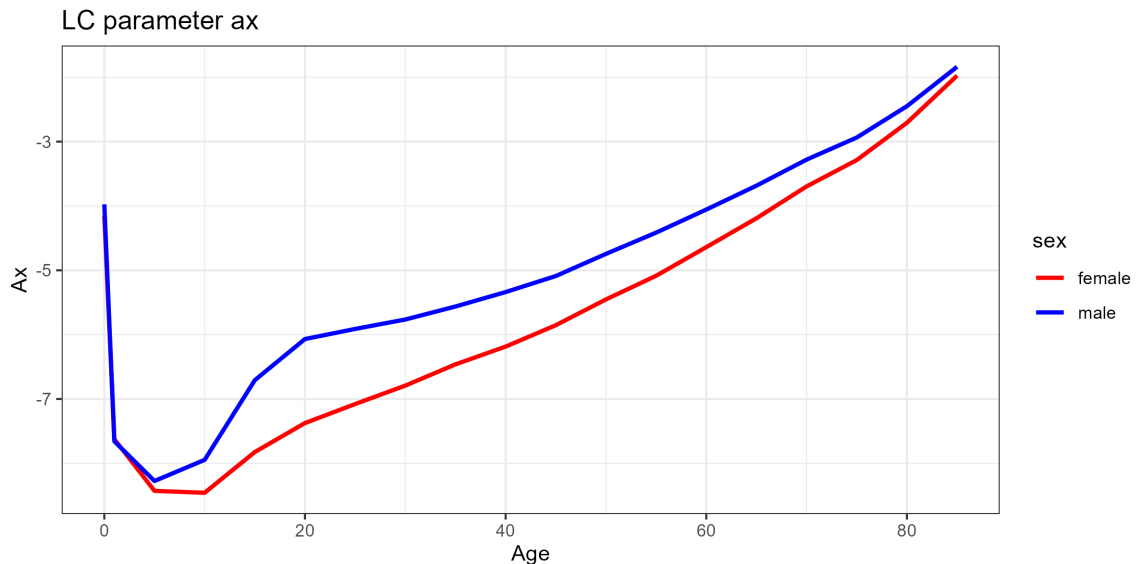
and extract the parameters. After several tests, we determined that 100 samples generated properly distributed residuals, but for robustness, we choose 500 to provide a more robust fit and still iterate at an acceptable pace. Following this we proceed to the simulations. The needed parameters for this are the number of simulations which we will set as the same as the number of bootstraps, the years to project set at 40. The jump choice, whether to use the fitted end or the actual end, we will choose the actual end in the data. Lastly for the simulation is the method of the projection of k_t . Our alternatives are multivariate random walk with drift and an N independent arima models. The result of of this is a series of 500 by 500 death rates for each of the 40 years to be projected for every age group. We can then select the projections that correspond to the median and the 97.5% and 2.5% quantiles to generate out confidence intervals for the death rates per age grou, the first of our final projects. Following this we can turn these into life tables to generate our final product, the life expectancy at birth projections with confidence intervals.

The initial fit

Fitted ax

Once the initial model fitting is completed, we have all our initial products. The first of these, a_x parameter, serves as the baseline mortality for the population. It represents the average log mortality rate for each age group across the entire time period. The a_x parameter captures the general shape of the mortality curve,

which typically shows high mortality rates at very young ages, a decline in mortality during childhood and early adulthood, and then an increase in mortality as people age. Ours graphs as follows.

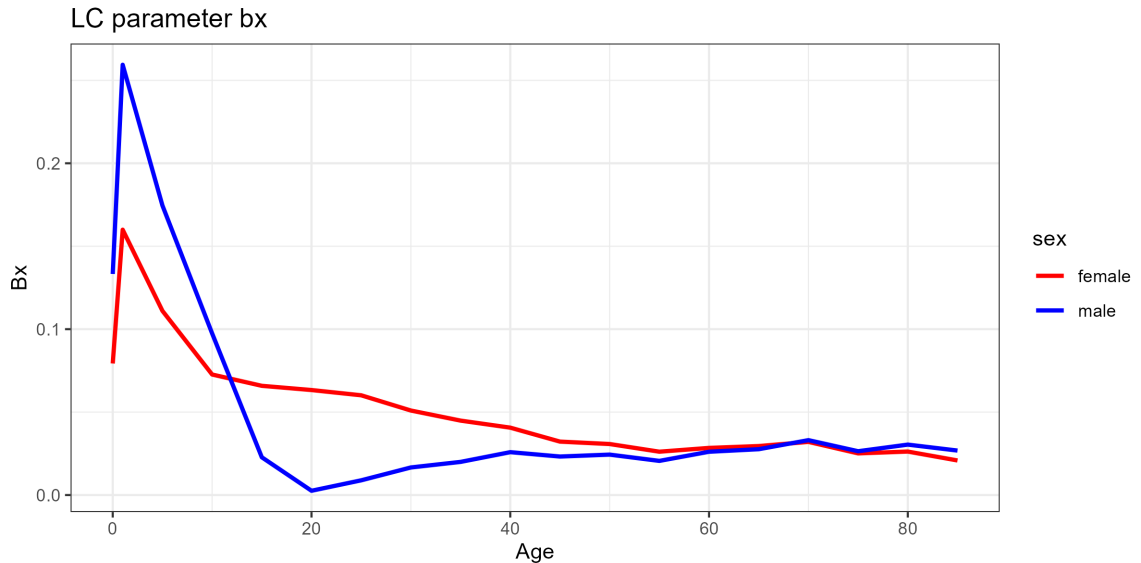


Here, we can see the drastic fall of the baseline log mortality after the first year of life. This is followed by the steady increase as one ages and the telltale male mortality hump starting at the teen years followed by a slow closing of the gap between the sexes as they age.

Fitted bx

Next we have the bx parameter, which is our age specific sensitivity to changes in the mortality index kt. The bx parameter captures how much the log mortality rate for each age group changes in response to changes in the mortality index kt. A higher bx value for a particular age group indicates that the mortality rate for that age group is more sensitive to changes in the overall mortality trend cap-

tured by kt. The bx parameter typically shows a pattern where certain age groups, such as infants and older adults, have higher sensitivity to changes in mortality trends, while other age groups, such as young adults, may have lower sensitivity. This reflects the fact that improvements in healthcare and living conditions can have a more significant impact on reducing mortality rates for certain age groups compared to others. In other words this parameter tends to be context specific and can vary significantly between populations and time periods. For our case, the bx parameter graphs as follows.

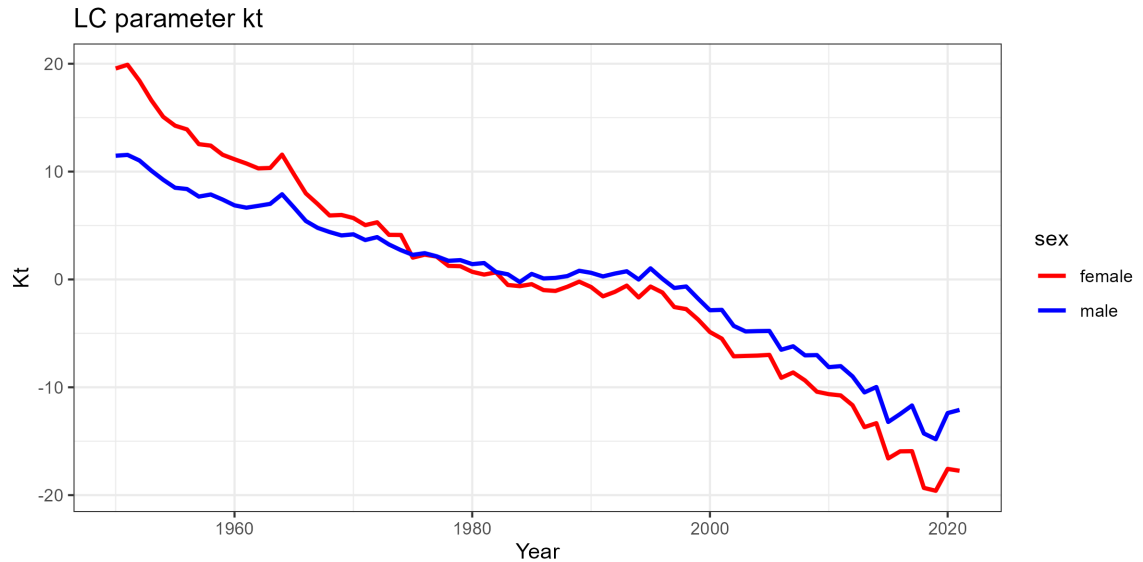


Here we can see the sensitivity of the mortality index to changes in mortality. The highest sensitivity is at early life. This is due to the monumental work done on the island in the latter half of the 20th century, better sanitation, better nutrition, and better healthcare. The high point is followed by an significant drop for both sexes at the teen years. Women then transition to a steady decline in sensitivity but males continue to drop until near zero at the age of 20 finally making a slow climb through to meet the female sensitivity at around 80 years of age. This is a

clear example of the context specific nature of the bx parameter, as the sensitivity of mortality to changes in the overall trend is not uniform across age groups and can be influenced by various factors such as healthcare improvements, lifestyle changes, and social determinants of health that affect different age groups differently. We will specifically see the effect of the parameter in the death rates and life expectancy projections later on, but for now we can move on to the kt parameter, which is our time-varying mortality index.

Fitted kt

The last of the initial fitted parameters, kt serves as the time-varying mortality index that captures the overall trend in mortality improvement over time. Historically, the kt parameter typically shows a declining trend over time, reflecting improvements in healthcare, living conditions, and other factors that contribute to reduced mortality rates. The kt parameter can also exhibit fluctuations due to various events such as epidemics, wars, or economic downturns that can temporarily increase mortality rates. For our case, the kt parameter graphs as follows.



The index corresponds to the average level of mortality for that year. There are fluctuations in the index, a spike in the mid 1960s, the effect of HIV in the mid 1990s (Calzada & Torres, n.d.) but the overall trend is a steady decline. The index starts off higher for women than men but they close the gap by the mid 1970 and start pulling apart as time passes leading to an slow growing gap, but both share the same general shape with the male and female indexes crossing in the mid 1970s.

Parameter simulation

Now that we have the initial fit, we can move on to the simulations. We provide the fitted parameters to the bootstrap method, which resamples the parameters to create new sets of parameters. These new parameters come with their own residuals to evaluate which we use to determine the number of bootstraps needed to create a robust distribution. Earlier, we stated that 500 samples were sufficient and these are the corresponding residuals.

ax residuals

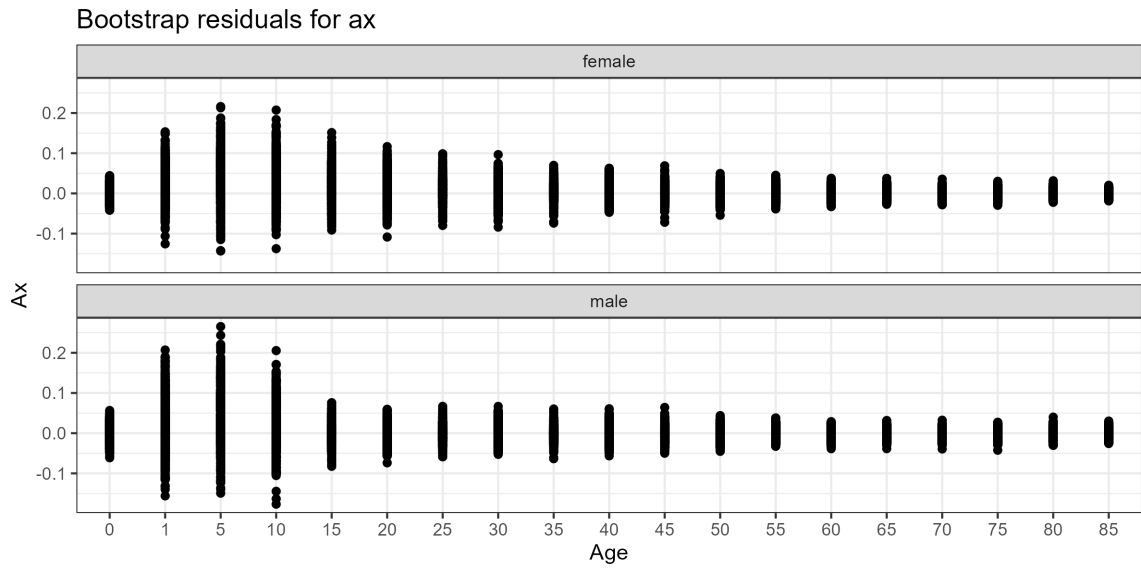


Table 1: Bootstrap residuals for ax

		Range		Median	Range		
	Median	Range	Difference	Fe-	Range	Difference	Range
Age	Male	Male	Male	male	Female	Female	Ratio
0	0.0004	(-0.0395 - 0.0369)	0.0764	0.0023	(-0.0304 - 0.032)	0.0625	0.818
10	0.0185	(-0.0934 - 0.1319)	0.2253	0.0348	(-0.0695 - 0.1407)	0.2102	0.933
20	-	(-0.0459 - 0.0443)	0.0902	0.0135	(-0.057 - 0.081)	0.1380	1.530
30	0.0001	(-0.039 - 0.044)	0.0830	0.0067	(-0.047 - 0.0596)	0.1066	1.285

		Range		Median	Range		
	Median	Range	Difference	Fe-	Range	Difference	Range
Age	Male	Male	Male	male	Female	Female	Ratio
40	-	(-0.0358 -	0.0737	0.0051	(-0.0339 -	0.0798	1.083
	0.0021	0.0378)			0.0459)		
50	-	(-0.0298 -	0.0552	0.0015	(-0.0297 -	0.0588	1.066
	0.0018	0.0254)			0.0292)		
60	-	(-0.0233 -	0.0452	0.0023	(-0.0229 -	0.0470	1.039
	0.0012	0.0219)			0.0241)		
70	-	(-0.0234 -	0.0431	0.0010	(-0.0199 -	0.0382	0.887
	0.0017	0.0198)			0.0184)		
80	-	(-0.0225 -	0.0440	0.0003	(-0.0169 -	0.0349	0.794
	0.0005	0.0215)			0.018)		

Here we have the ax residuals of the bootstrap. It presents a large deviance spike around childhood. For males this tapers quickly when entering the teen years and continues to narrow through adulthood and the senior years. Women on the other hand present a slower narrowing into adulthood and further slows down past the 40 year mark. Looking at the quantiles, at the median there's a clear skew towards opposite sides of zero between the sexes and the range between the 95% quantiles between men and woman are withing 20% of each other. In short both share the same general shape of uncertainty with the males having a slight offset towards the negative. This finally leads to males having a lower baseline level of

log mortality than females

bx residuals

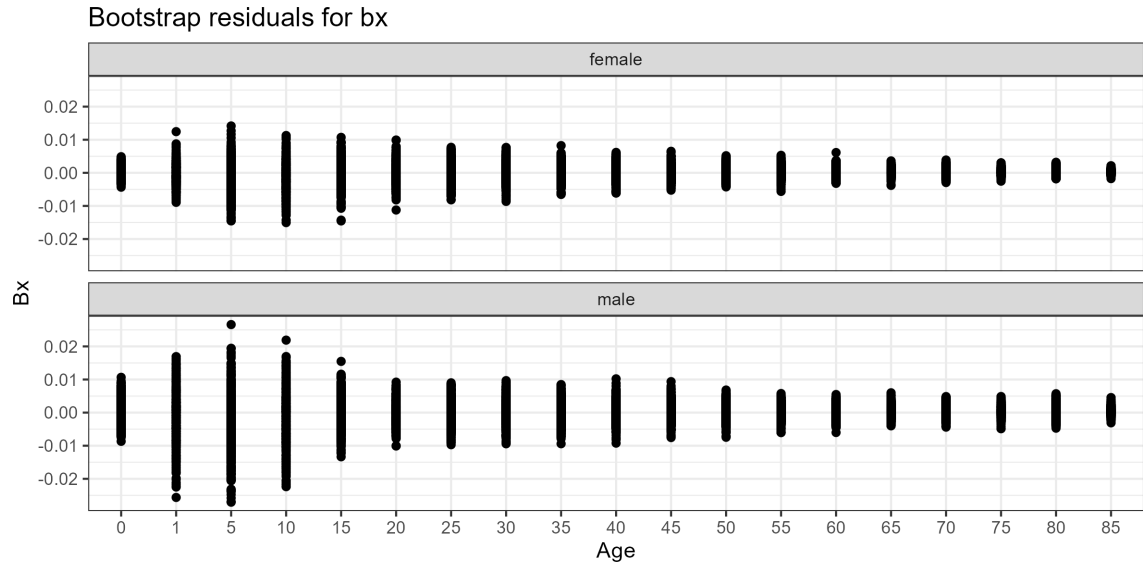


Table 2: Bootstrap residuals for bx

		Range		Median	Range		
	Median	Range	Difference	Fe-	Range	Difference	Range
Age	Male	Male	Male	male	Female	Female	Ratio
0	0.0009	(-0.0057 - 0.0077)	0.0134	0.0004	(-0.0027 - 0.0034)	0.0061	0.451
10	-0.0012	(-0.0178 - 0.0142)	0.0320	-0.0014	(-0.0105 - 0.0074)	0.0180	0.562
20	0.0000	(-0.0067 - 0.0071)	0.0137	-0.0003	(-0.0061 - 0.0061)	0.0123	0.893

		Range		Median	Range		
	Median	Range	Difference	Fe-	Range	Difference	Range
Age	Male	Male	Male	male	Female	Female	Ratio
30	0.0001	(-0.0069 - 0.0071)	0.0140	0.0003	(-0.0047 - 0.0057)	0.0104	0.740
40	0.0004	(-0.0047 - 0.0061)	0.0108	0.0006	(-0.0043 - 0.0047)	0.0090	0.831
50	0.0004	(-0.004 - 0.0048)	0.0089	0.0004	(-0.0031 - 0.0036)	0.0067	0.757
60	0.0005	(-0.0034 - 0.0039)	0.0074	0.0003	(-0.0022 - 0.0031)	0.0053	0.714
70	0.0006	(-0.0027 - 0.0039)	0.0065	0.0005	(-0.0016 - 0.0024)	0.0040	0.607
80	0.0004	(-0.0027 - 0.0039)	0.0067	0.0003	(-0.0014 - 0.0023)	0.0037	0.557

Moving on to the bx parameters bootstrap residuals, in many ways it mirrors the ones from the ax parameter, with a large deviance spike around childhood that tapers off as age progresses. However, unlike the ax residuals, we see a clear difference in the range of the residuals between sexes. For males, the residuals are much more spread out than the female counterpart, with the 95% range ratio being ranging between 0.50 and 0.94, meaning that the male sensitivity to changes in the mortality index is much more volatile than the female sensitivity.

kt residuals

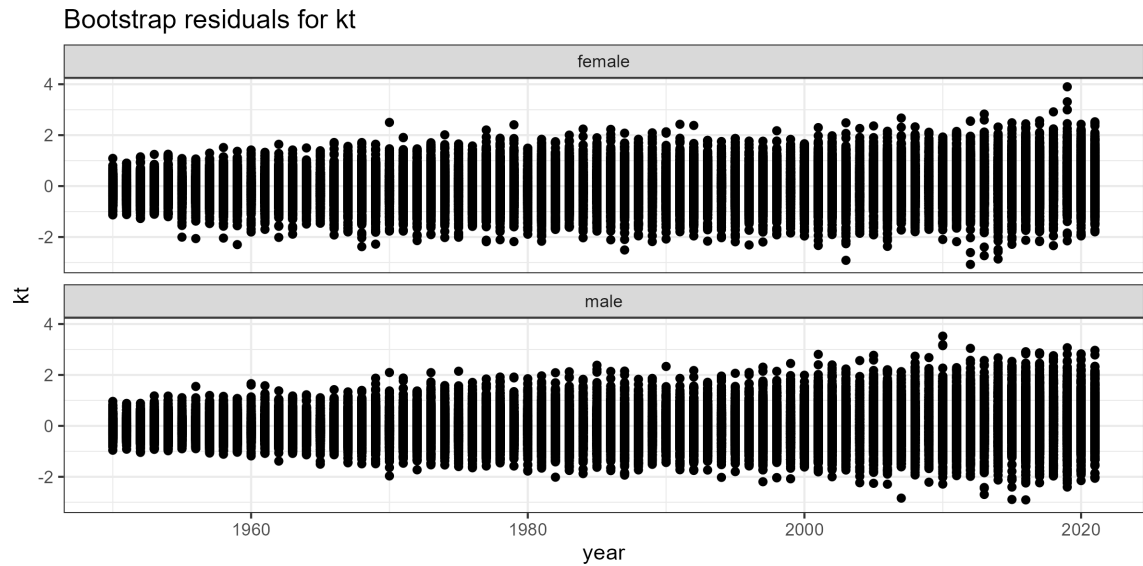


Table 3: Bootstrap residuals for kt

		Range		Median	Range		
	Median	Range	Difference	Fe-	Range	Difference	Range
Year	Male	Male	Male	male	Female	Female	Ratio
1950	-	(-0.6227 -	1.1991	-	(-0.766 -	1.2526	1.045
	0.1061	0.5763)		0.0984	0.4866)		
1960	-	(-0.7559 -	1.5383	-	(-1.0584 -	1.8355	1.193
	0.0927	0.7824)		0.1223	0.7771)		
1970	-	(-0.982 -	1.9699	-	(-1.2665 -	2.2814	1.158
	0.0930	0.9879)		0.0996	1.0149)		
1980	-	(-1.1086 -	2.2962	-	(-1.4205 -	2.6827	1.168
	0.0273	1.1876)		0.0788	1.2622)		

Year	Median		Range		Median		Range	
	Male	Female	Male	Female	Male	Female	Male	Female
1990	-0.0917	-0.0917	-1.131	-1.131	2.2624	2.2624	-1.3342	-1.3342
2000	-0.0070	-0.0070	-1.2377	-1.2377	2.4904	2.4904	-1.2817	-1.2817
2010	0.0761	0.0761	-1.328	-1.328	3.0812	3.0812	-1.3828	-1.3828
2020	0.2301	0.2301	-1.5447	-1.5447	3.4322	3.4322	-1.1854	-1.1854

Moving on to the residuals for the kt variable, we see that unlike both other parameters, kt slowly fans out over time. We also see the medians are likewise, slowly climbing as time progresses unlike the two other parameters and this is consistent for both sexes. Meanwhile the range ratio between the sexes is shifting first skewed towards females but now centers around 0.92.

The forecast

Now, that we have the bootstrap parameters, we can now do the simulations. The simulations are done by taking the bootstrapped parameters and simulating the kt variable forward in time. This allows us to create a distribution of possible future

mortality scenarios based on the uncertainty in the parameters. The results of the simulations can be used to create confidence intervals for the future mortality rates. The parameters that define the simulations are the number of simulations, the years to project, the jump choice, and the method of projection for k_t . For our case, we set the number of simulations to 500, the same as the bootstrap samples, the years to project, 40 in our case, the jump choice, the actual end in our data. The method of projection for k_t , either multivariate random walk with drift or iarima models, the vast majority of literature uses the random walk and we have no reason to deviate from this (Girosi & King, 2007).

Simulated k_t

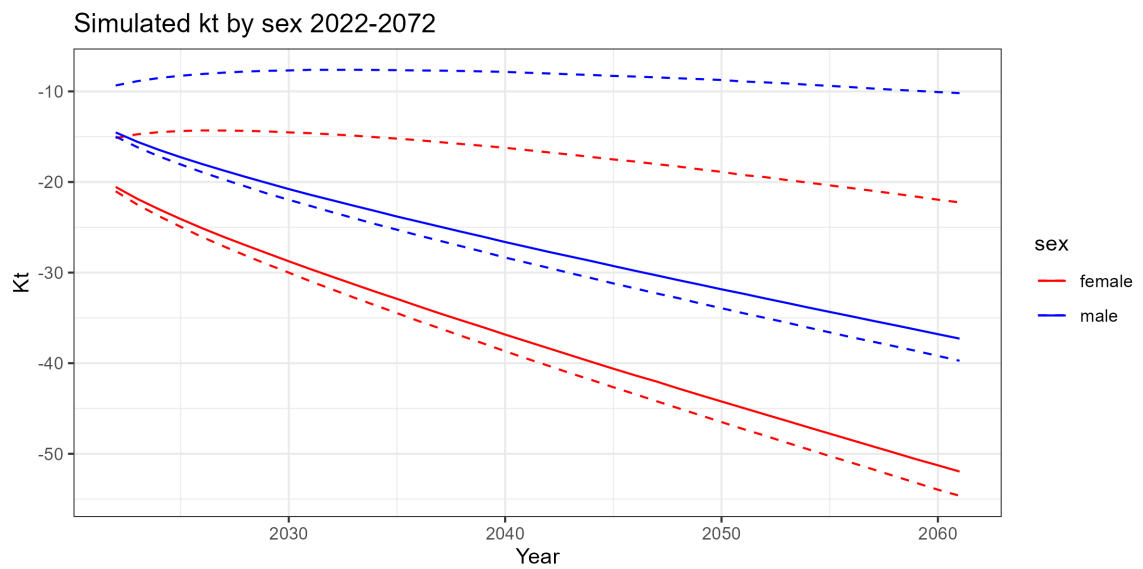


Table 4: Kt simulations 2022-2050.

		Range		Median		Range	
Year	Median	Range	Difference	Fe-	Range	Difference	Range
	Male	Male	Male	male	Female	Female	Ratio
2030	-	(-21.9653 -	14.2748	-	(-30.0018 -	15.4771	1.084
	20.7774	-7.6905)		28.7489	-14.5247)		
2040	-	(-28.3392 -	20.4930	-	(-38.6742 -	22.4492	1.095
	26.6265	-7.8461)		36.8394	-16.225)		
2050	-	(-33.9282 -	25.1865	-	(-46.4961 -	27.6175	1.097
	31.8454	-8.7417)		44.2246	-18.8786)		
2060	-	(-39.1934 -	29.1208	-	(-53.9604 -	32.0074	1.099
	36.7956	-10.0726)		51.2680	-21.9529)		

Now, looking at the projections for the kt variable, we see that both medians fall at a steady rate over time. As expected of the random walk both ranges spread out over time, but the range ratio is steady between sexes and roughly 1.1. Something of note is the behavior of the projected median. It almost falls to the lower bound for both sexes, which is a sign of a skewed distribution.

Results

Death rates

Now, with the kt projections, we can now calculate the first of our two final products, the projected mortality rates for each age and sex by using the ax and bx parameters. The projected mortality rates can be calculated using the formula: $\log(mortality_{rate}) = ax + bx * kt$. This allows us to create a distribution of possible future mortality rates based on the uncertainty in the parameters and the projections of kt . The projected mortality rates can be used to create confidence intervals for the future mortality rates.

Table 5: Death Rates Historic and Forecasted 2020-2050.

age	year	central	lower	upper	type	sex
20	2000	-7.7218	NA	NA	historic	female
20	2025	-8.0829	-7.7639	-8.4041	forecast	female
20	2050	-8.9031	-8.0403	-9.7823	forecast	female
40	2000	-6.3572	NA	NA	historic	female

age	year	central	lower	upper	type	sex
40	2025	-6.6742	-6.4669	-6.8835	forecast	female
40	2050	-7.2080	-6.6464	-7.7833	forecast	female
60	2000	-4.8466	NA	NA	historic	female
60	2025	-5.1073	-4.9620	-5.2534	forecast	female
60	2050	-5.4810	-5.0881	-5.8808	forecast	female
20	2000	-6.0309	NA	NA	historic	male
20	2025	-6.1839	-6.1665	-6.2139	forecast	male
20	2050	-6.1972	-6.1354	-6.3123	forecast	male
40	2000	-5.4020	NA	NA	historic	male
40	2025	-5.3750	-5.2545	-5.4990	forecast	male
40	2050	-5.5879	-5.2651	-5.9307	forecast	male
60	2000	-4.1723	NA	NA	historic	male
60	2025	-4.3573	-4.2359	-4.4800	forecast	male
60	2050	-4.5745	-4.2453	-4.9088	forecast	male



Here we have one of our final products, the death rates. Using the median projection and dividing by age group and sex, we see that minors in both sets are projected to have a continued accelerated mortality decline through the projected period though their mortality is highly erratic especially by the end of our historic data. Adult women are also projected to have a continued decline in mortality but at a slower pace than minors, while mortality is projected to plateau for young adult. Those that survive to late adulthood are projected to continue their decline in mortality but at a much slower pace than that of women. Finally for both senior groups we see a continued decline in mortality with women projected to have a much faster decline. One of the more important things to note is that the original bx parameter describes much of this graph. The high sensitivity of the minors to changes in the mortality index kt means that they are projected to have a much faster decline in mortality than the other age groups. The low sensitivity of the young adults

means that their mortality is projected to plateau, while the moderate sensitivity of the adults and seniors means that they are projected to have a continued decline in mortality but at a slower pace than minors. Meanwhile the ax parameter is overwritten by our explicitly requirement to force the projections to start at the actual end of the data, so it does not have a direct effect on the projections but it does have an indirect effect by influencing the initial fit of the model and therefore the parameters that are used in the simulations.

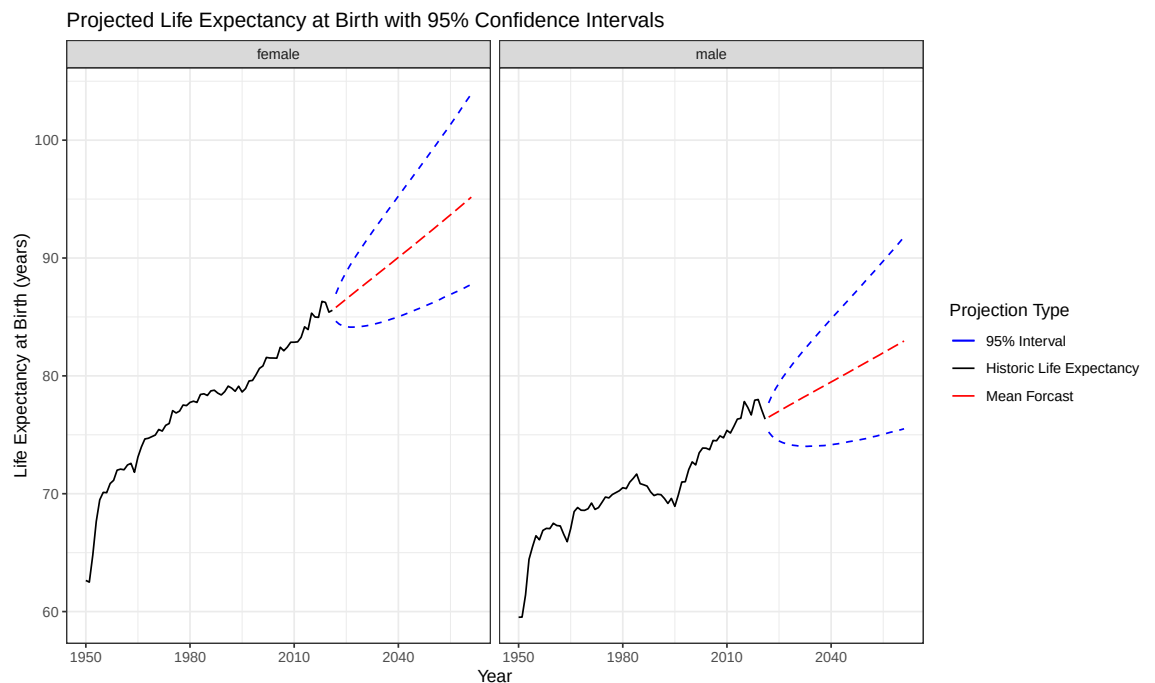
Life Tables and Life expectancy at birth

Now for the second final product. Proper life tables for both sexes and the life expectancy at birth projected through the entire period.

Table 6: Life Tables historic and Median Projection 2000-2060.

sex	year	age	n	qx	lx	dx	Lx	Tx	ex	type
female	2000	0	1	0.0091	1e+05	907	99156.09	806335280.6335		historic
female	2010	0	1	0.0066	1e+05	660	99381.47	828558682.8559		historic
female	2020	0	1	0.0077	1e+05	771	99279.76	854075185.4075		historic
female	2030	0	1	0.0046	1e+05	456	99570.13	877023187.7023		central
female	2040	0	1	0.0030	1e+05	301	99714.99	900542490.0542		central
female	2050	0	1	0.0020	1e+05	199	99811.02	924393892.4394		central
female	2060	0	1	0.0013	1e+05	132	99874.41	949075294.9075		central
male	2000	0	1	0.0108	1e+05	1083	98997.54	727017572.7017		historic

sex	year	age	n	qx	lx	dx	Lx	Tx	ex	type
male	2010	0	1	0.0085	1e+05	854	99204.16	753744	775.3745	historic
male	2020	0	1	0.0065	1e+05	653	99387.89	771188	177.1188	historic
male	2030	0	1	0.0054	1e+05	537	99494.94	778433	177.8433	central
male	2040	0	1	0.0035	1e+05	346	99672.79	794817	879.4818	central
male	2050	0	1	0.0022	1e+05	223	99788.37	811277	981.1278	central
male	2060	0	1	0.0014	1e+05	144	99863.04	827934	182.7934	central



Seeing the life expectancy at birth projections, we see that the historic data shows a steady increase in life expectancy. The projected values show this trend with a continued increase in life expectancy of about 2.3 years per decade for women and 1.6 years per decade for men. The median of the projection reaches 94 years for women and 82 for men. The confidence intervals on the other hand

present a curious picture. The projection intervals are large enough to warrant a closer look, in particular, the upper bound of the female life expectancy projection presents linear growth and is projected to reach 104 years by 2060 and male life expectancy reaches 92 year. The lower bound for both seem much more plausible, with the female projection reaching 89 years and the male 76.

Discussion

Our benchmarks

We have already stated that there are no modern projections of mortality and life expectancy projections that are specific to Puerto Rico, so we will be comparing our results to three entities that do produce projections for the entire world, including Puerto Rico: the UN World Population Prospects (WPP), the U.S. Census Bureau's International Data Base (IDB), and the University of Washington's Institute for Health Metrics and Evaluation (IHME). Each of these entities uses different methodologies and assumptions, so comparing our results to theirs will provide a useful context for understanding the implications of our projections.

The WPP projects mortality using a two-phase probabilistic model that combines historical trends with country-specific data and model life tables. It begins by projecting life expectancy at birth using a double-logistic function, capturing rapid improvements in early mortality followed by slower gains in older ages. For data-rich countries, the Lee-Carter method is applied to age-specific mortality rates, while for data-scarce countries, model life tables (e.g., Coale-Demeny or UN fami-

lies) are calibrated using indicators like under five mortality and adjusted via the Calibrated Spline method. Regional benchmarks temper extreme trends such as rapid gains in fragile states or stagnation in advanced nations ensuring plausible long-term projections. The WPP also incorporates the impact of crises like pandemics and conflicts through excess mortality estimates, and uncertainty is quantified using prediction intervals. These mortality projections are then used in cohort-component models to forecast population change.

The IDB produces mortality and life expectancy projections for Puerto Rico and other global regions using a multi-step cohort-component methodology. Base mortality estimates are derived from high-quality data sources like vital registration systems, surveys and censuses, with adjustments for underenumeration, age heaping, and data quality issues. For countries with limited data, model life tables are used to estimate age and sex-specific mortality patterns. The IDB extends mortality projections to age 100+ using logistic functions to ensure smooth, realistic trends, even in the absence of empirical data. For countries affected by HIV/AIDS, the IDB employs a dual-scenario approach—modeling mortality under both “with AIDS” and “without AIDS” conditions using data from the UNAIDS Spectrum software to incorporate HIV prevalence and epidemic dynamics. Projections for future mortality are based on logistic curves fitted to historical life expectancy trends, with fixed slope models ensuring consistent long-term improvements, capped at 95 years for males and 100 for females by 2100. These projections are validated against demographic indicators and adjusted for shocks like wars, famines, and natural disasters, ensuring high precision in capturing the timing and impact of major events.

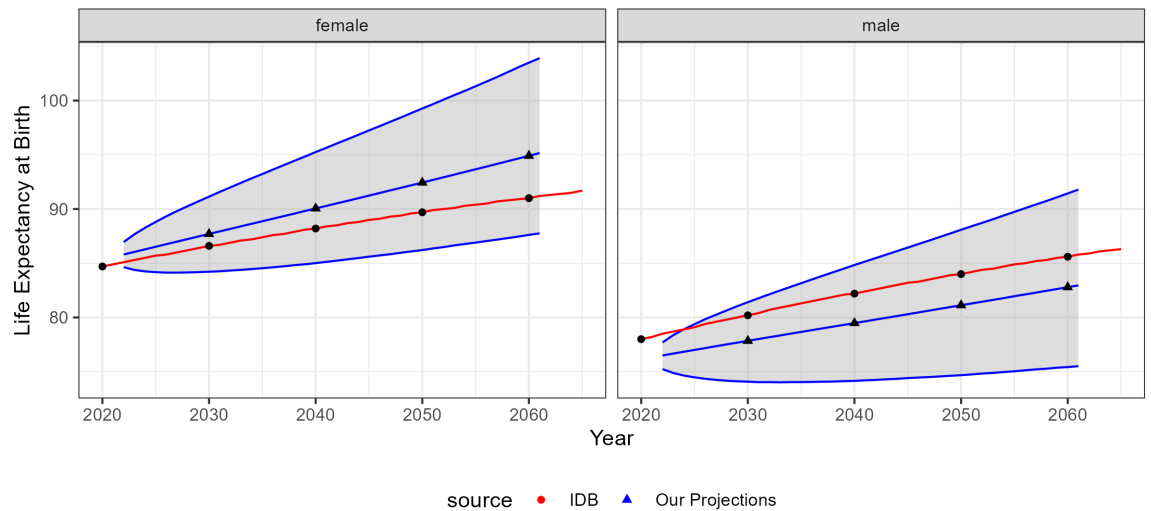
Finally, the IHME projects mortality using a data driven model that integrates the Socio-demographic Index, cause-specific risk factors from the Global Burden of Diseases study, and an ARIMA component to capture unexplained mortality trends. The model estimates underlying mortality as a function of SDI, time, and risk factors, while explicitly accounting for the interplay of risk factors such as tobacco, obesity, and air pollution. An ARIMA model with attenuated drift is used to project residual mortality, ensuring robust long-term forecasts without overfitting. This approach enables IHME to generate probabilistic, scenario-based projections for mortality and population, incorporating uncertainty across all components and providing a dynamic, causal framework distinct from more deterministic models used by other organizations.

Comparison with External Benchmarks

With our actors and their methodologies defined, we can now compare our projections to theirs. The nature of the International Databases projections mean that they do not produce confidence intervals, so we will be comparing our median projections to their point estimates. In the graph below, we can see that our median projections fall on opposite ends of the IDB. Our female mortality projection starts slightly above the IDBs point estimate and then diverges upwards slowly, while our male mortality starts below the IDB point estimate point and keeps the gap until the end of the projection period. In short, females will outpace the IDB projections while the males will consistently lag behind the IDB projections. In both cases, the

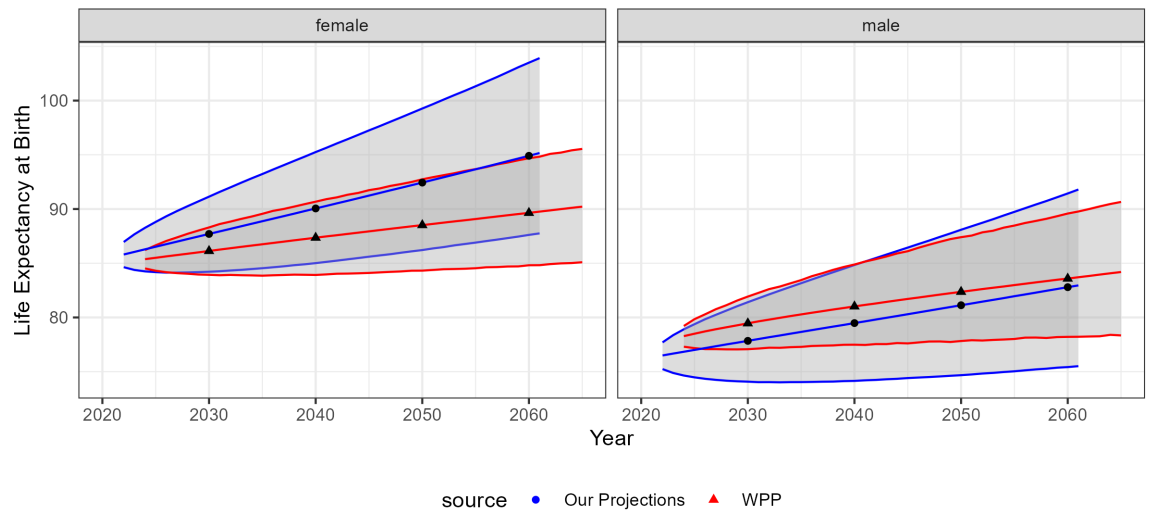
IDB projections lie within our confidence intervals.

Comparison of our projections with the IDB: 2023:2062



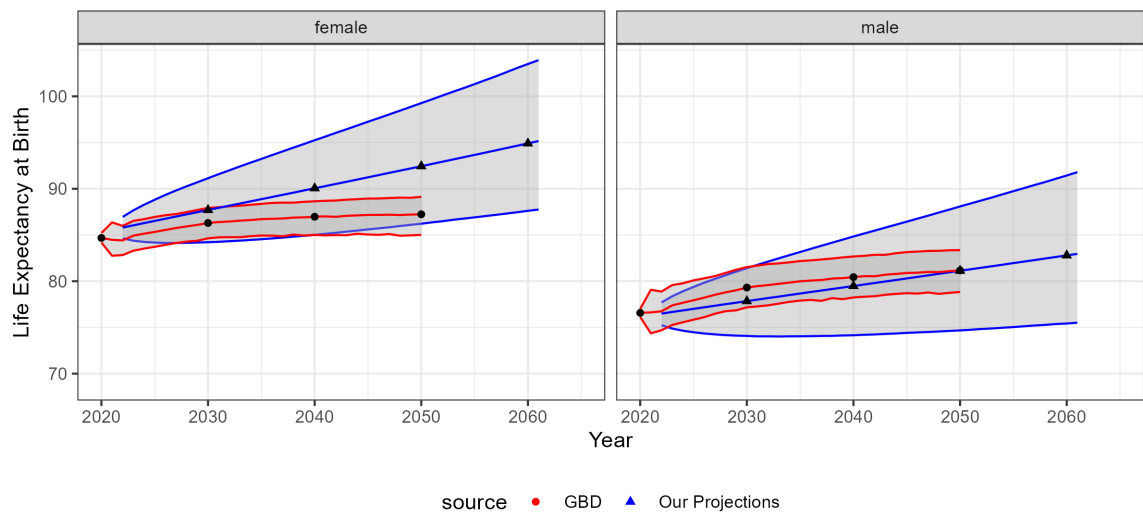
With the WPP, we can see a different pattern. In our case, our median line for female life expectancy starts above the WPP median but instead of tracking the WPP median, it diverges upwards tracking the WPP upper bound, while our female upper bound diverges even more upwards. For males, the story is more plausible, with our male median starting below the WPP median but climbs faster than the WPP median. The upper bound of our male projections tracks the WPP upper bound until the second half of the projection period where it diverges upwards, while the lower bound is always below the WPP lower bound. This is a more plausible pattern.

Comparison of our projections with the WPP: 2023:2062



Finally, the IHME projections are much more difficult to compare to our projections. They do provide confidence intervals, but they don't produce the typical fanning out of the confidence intervals that we would expect from a stochastic model. Instead, their confidence intervals are much more narrow and they don't show the same pattern of divergence that we see in our projections. Their model implies that there will be a plateauing of life expectancy for both males and females in the 2040s, while every other projection, including ours, shows a continued increase in life expectancy through the end of the projection period. This is a significant divergence from our projections and the other benchmarks. That being said, the IHME projections generally lie within our confidence intervals, though our medians behave very differently from the IHME medians, with our medians showing a continued increase in life expectancy while the IHME medians show a plateauing of life expectancy. This is a significant divergence from our projections and the other benchmarks.

Comparison of our projections with the GBD: 2023:2062



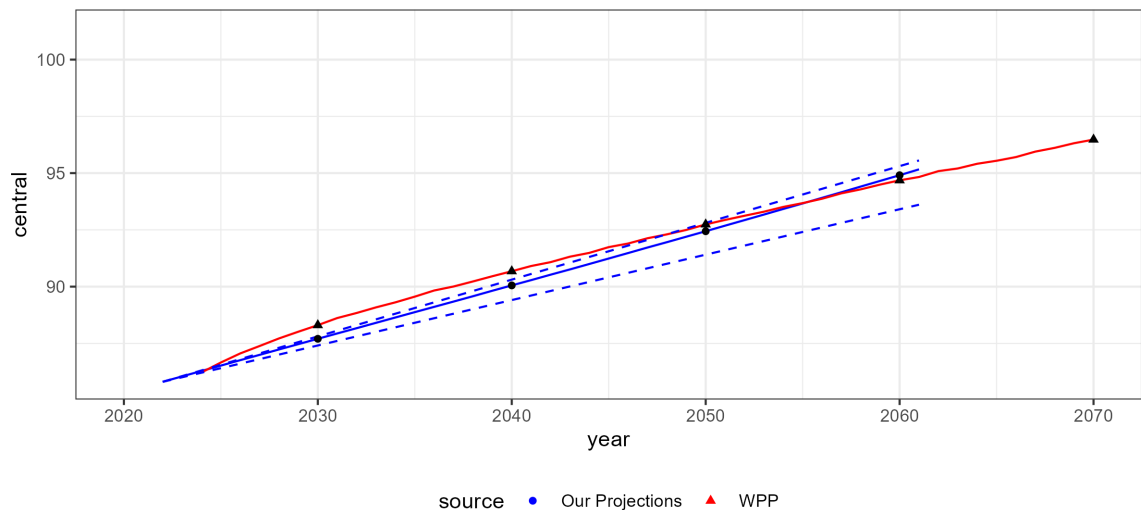
When we see the then median of the male projections for the WPP and IDB, we see that they are quite close to our median projections, with our projections for 2061 running just below the WPP at 82 years the WPP at 83 years and the IDB at 85 years. Our intervals do behave differently with ours presenting much more uncertainty in the lower bound growing much faster than the WPP then curving back up, until the end of the projection period where it stops at 75 years while the WPP's lower bound reaches 78 years. The upper bound of our projections roughly tracks the WPP until it starts diverging upwards after 2045, reaching 91 years by 2061 while the WPP's upper bound reaches 89 years.

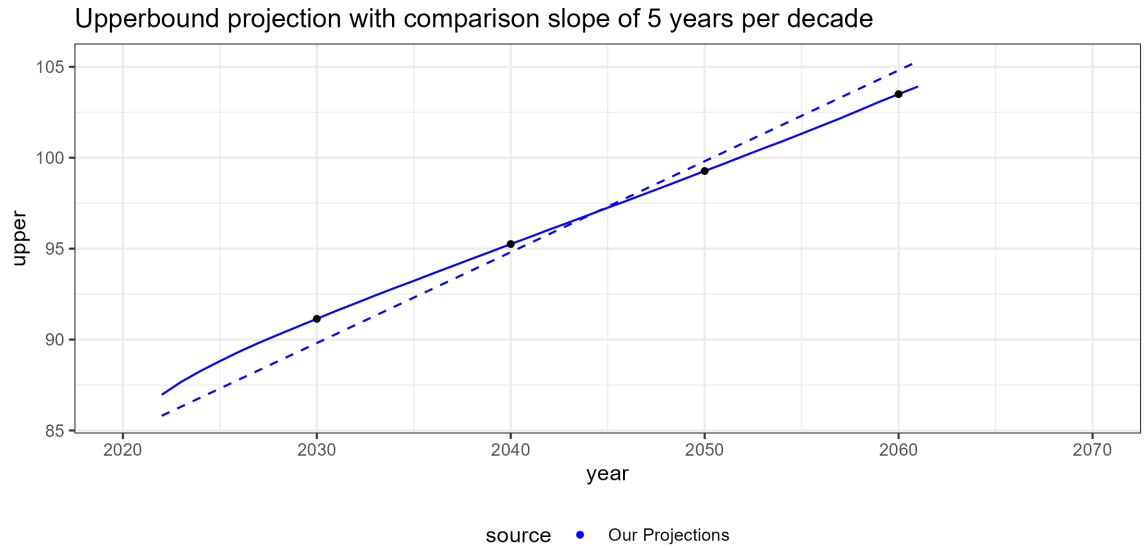
The male projections have a clear downwards drift on the lower bound at the beginning of the projection period, but it prescribes a plausible trajectory given the much more erratic nature of male mortality i.e. it's not shocking. What is shocking is the behavior of the female projections, where the median projection passes the upper bound of the WPP projections by the end of the projection period with

ours reaching 94 years while the WPP reaches 90 years. The lower bound of our projections reaching 95 years with the IDB presenting 91 and the WPP 89. The upperbound for our projections reaches 103 years while the WPP reaches 94 years. and the lower bound of our projections reaches 87 years while the WPP reaches 84 years.

Oeppen and Vaubel stated in may 2002 that “best-practice life expectancy gains” have been about 2.5 years per decade(Oeppen & Vaupel, 2002), meanwhile Ronald Lee states its likely closer to 2.0 years per decade (R. Lee, 2002). If we graph those limits, we can see that our median projection for women exceeds the 2.0 year per decade limit stated by Ronald Lee, and approaches the 2.5 year per decade limit stated by Oeppen and Vaubel. This isn’t even considering the upper bound of our projections, which exceeds the 2.5 year per decade limit by a wide margin, reaching 5.0 years per decade as we show below

Comparison of our projections with WPP upper bound and projection median with theoretical





The Skewing of female mortality

It's well known that the Lee-Carter model can produce skewed projections for life expectancy, particularly for longer projection horizons, due to the way uncertainty in the k_t index translates into life expectancy. In our projections for Puerto Rico, this skew is particularly pronounced and we have done some analysis to understand why that is the case. The usual suspects have been ruled out, as the model fit is good, the b_x parameters are reasonable and the historical k_t index does not show any unusual patterns that would suggest a structural break. Following those checks we've double checked the code, both verifying the implementation of the Lee-Carter model and the random walk with drift projection of k_t , we've iterated through all the parameters to see if any of them could be driving the skew. In fact, the parameters themselves do little to change the structural shape of the projections, with both the median and the upperbound of female mortality showing

unrealistic improvements in life expectancy at birth, while the lower bound shows a slightly more plausible trajectory under all scenarios.

Model Performance and Limitations

Pleateau in young males

It is well known that the young adult male population in Puerto Rico has been affected by high rates of violence and accidents, which has led to excess mortality in that age group. The Lee-Carter model captures this historical pattern through the b_x parameters, which shows a very low sensitivity to changes in the k_t . Most of the enviromental change over the past decades have been in areas of healthcare and public health. It is likely that these changes have been canceled out by the excess mortality from violence and accidents, leading to a plateau in the projections for young males. But this is a known limitation of the Lee-Carter model, which does not account for changes in the underlying causes of mortality or for structural breaks in the data. The model assumes that the age-specific mortality rates will continue to evolve in a similar way to the past, which may not be the case for Puerto Rico given the unique challenges it faces.

Future shocks

The model was able to properly capture the mortality shock of the mid 90s from HIV. However, the random walk with drift projection of the k_t index does not

allow for the anticipation of future shocks, which is a known limitation of the Lee-Carter model. This is particularly relevant for Puerto Rico, which has a history of hurricane-driven excess mortality. The model may not be able to capture the potential impact of future hurricanes or other natural disasters on mortality rates, which could lead to an overestimation of life expectancy in the long run.

Broader Implications

Even if our projections outperform the benchmarks by an unrealistic margin, the data tells us two things. First, that there is a significant amount of uncertainty in the projections, particularly for females and particularly in the upper limits of mortality, which is reflected in the wide confidence intervals. This uncertainty is not just a modeling limitation but a reflection of the complex and dynamic nature of mortality trends in Puerto Rico, which are influenced by a wide range of factors including healthcare, public health, social and economic conditions, environmental and migratory factors. Second, that life expectancy in Puerto Rico is likely to continue to improve in the future, but the rate of improvement is uncertain and may be influenced by a wide range of factors that are difficult to predict. This has important implications for policymakers and planners, who need to be prepared for a range of possible scenarios when it comes to healthcare and social services for the aging population in Puerto Rico. The wide confidence intervals themselves are a finding — they communicate to policymakers that demographic uncertainty in Puerto Rico is not merely a modeling limitation but a genuine planning risk that demands

scenario-based fiscal and healthcare planning rather than single-point forecasts.

Further more a deeper look into the source of the skew is needed to understand the underlying drivers of mortality trends in Puerto Rico and to improve the robustness of our projections. The two main data sets, local census estimates for population and the local health department for vital statistics, are the likely sources of the skew. The census might be overestimating the population on the latest decennial census which was done during the pandemic. But this would mean there is a significant over counting of the population, leading to a much lower mortality rate over the past decade and the years that follow. There have been no vital statistics reports in the past years, but the health department reports to the national vital statistics system, so it is unlikely that there has been a significant under counting of deaths since we were not filtering for any specific cause of death. Lastly the boogiemans in demography in Puerto Rico is the out migration over the past decade. But for this to be the cause it requires that the population leaving specifically have a higher likelihood of dying than the population that remains, which is possible but we have no evidence to support nor believe this claim. Further research is needed to understand the source of the skew and to improve the robustness of our projections, but it is important to note that the skew is not a modeling artifact but rather a reflection of the data and the events that have shaped mortality in Puerto Rico in recent years.

Recomendations

The underlying model we used, the Lee-Carter model, is a widely used and well-established model for projecting mortality and life expectancy. Having said that, it's clear that the model has limitations, particularly in the context of Puerto Rico where there are unique challenges and structural breaks in the data. As such our recommendation is to utilize the hierarchical Bayesian model developed by the United Nations World Population Prospects (Raftery et al., 2013), which incorporates a much larger context of data and events, and is likely to be more robust to the kinds of structural breaks that we have identified in our analysis. Having said that, the Lee-Carter model and this analysis still useful. The hierarchical Bayesian model is a much more complex model that requires a much larger amount of data and computational resources. While it does generate much more believable projections, one would not be able to identify the structural breaks in the data that we have identified with the Lee-Carter model, which is a useful finding in and of itself. The Lee-Carter model is also much more transparent and easier to understand and interpret, which can be useful for communicating the results to policymakers and stakeholders.

Future work

The next step in this project is to implement the hierarchical Bayesian model developed by the United Nations World Population Prospects (Raftery et al., 2013) and compare its projections to those generated by the Lee-Carter model. Additionally, the source of the skew in the projections needs to be further investigated, particularly in relation to the data from 2010 to 2023. This could involve a deeper analysis of the events that occurred during that period and how they may have impacted mortality trends in Puerto Rico, as well as an examination of the data collection methods and quality during that time. Finally, it would be useful to explore other modeling approaches that can account for structural breaks and changes in the underlying causes of mortality. This could include models that incorporate covariates related to healthcare, public health, social and economic conditions, environmental and migratory factors as we have not included any covariates in our projections. Finally, these projections should be used to inform scenario-based planning for healthcare and social services for the aging population in Puerto Rico, taking into account the wide range of possible outcomes and the significant uncertainty in the projections as well as serve as a stand-in for the more complex

hierarchical Bayesian model until we are able to implement it.

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Appendix

Tables

Summary of Death Rates

Table 7: Death Rates Puerto Rico Historic: 1950-2021.

year	age	male	female	year	age	male	female
1950	0	0.0828	0.0672	2010	0	0.0086	0.0066
	20	0.0042	0.0036		20	0.0026	0.0003
	40	0.0076	0.0070		40	0.0031	0.0014
	60	0.0212	0.0182		60	0.0127	0.0068
	80	0.0959	0.0771		80	0.0672	0.0488
2000	0	0.0109	0.0092	2020	0	0.0066	0.0078
	20	0.0024	0.0004		20	0.0016	0.0004
	40	0.0045	0.0017		40	0.0037	0.0013
	60	0.0154	0.0079		60	0.0127	0.0062
	80	0.0779	0.0599		80	0.0554	0.0370

Table 8: Death Rates Puerto Rico Projections:Median 2022-2062.

year	age	male	female	year	age	male	female
2022	0	0.0077	0.0064	2050	0	0.0023	0.0020
	20	0.0021	0.0003		20	0.0020	0.0001
	40	0.0047	0.0013		40	0.0037	0.0007
	60	0.0132	0.0063		60	0.0103	0.0042
	80	0.0563	0.0381		80	0.0427	0.0259
2030	0	0.0054	0.0046	2062	0	0.0013	0.0012
	20	0.0021	0.0003		20	0.0020	0.0001
	40	0.0044	0.0011		40	0.0034	0.0006
	60	0.0123	0.0056		60	0.0093	0.0035
	80	0.0520	0.0342		80	0.0379	0.0220

Table 9: Death Rates Puerto Rico Projections:97.5 Percentile 2022-2062.

year	age	male	female	year	age	male	female
2022	0	0.0056	0.0052	2050	0	0.0004	0.0007
	20	0.0020	0.0003		20	0.0018	0.0001
	40	0.0045	0.0012		40	0.0026	0.0004
	60	0.0124	0.0059		60	0.0073	0.0028
	80	0.0524	0.0357		80	0.0293	0.0182
2030	0	0.0022	0.0025	2062	0	0.0002	0.0004
	20	0.0020	0.0002		20	0.0018	0.0000
	40	0.0037	0.0008		40	0.0022	0.0003
	60	0.0103	0.0045		60	0.0061	0.0022
	80	0.0425	0.0276		80	0.0239	0.0146

Table 10: Death Rates Puerto Rico Projections:2.5 Percentile 2022-2062.

year	age	male	female	year	age	male	female
2022	0	0.0104	0.0078	2050	0	0.0125	0.0059
	20	0.0021	0.0004		20	0.0021	0.0003
	40	0.0050	0.0015		40	0.0052	0.0013
	60	0.0140	0.0068		60	0.0145	0.0062
	80	0.0603	0.0409		80	0.0628	0.0372
2030	0	0.0135	0.0082	2062	0	0.0098	0.0043
	20	0.0021	0.0004		20	0.0021	0.0002
	40	0.0053	0.0015		40	0.0050	0.0011
	60	0.0148	0.0069		60	0.0138	0.0055
	80	0.0641	0.0415		80	0.0593	0.0334

Summary of Lifetables

Table 11: Life Table Male Historic 1950-2021.

year	age	n	qx	lx	dx	Lx	Tx	ex
1950	0	1	0.0780	100000	7805	94280.36	5951951.7	59.5195
	20	5	0.0208	86502	1801	428007.50	4184276.7	48.3720
	40	5	0.0371	77683	2882	381210.00	2535584.2	32.6401
	60	5	0.1008	62250	6273	295567.50	1118061.7	17.9608
	80	5	0.3866	26380	10199	106402.50	201211.7	7.6274
2000	0	1	0.0108	100000	1083	98997.54	7270174.7	72.7017
	20	5	0.0119	97944	1170	486795.00	5297639.8	54.0885
	40	5	0.0223	92540	2063	457542.50	3387889.8	36.6100
	60	5	0.0742	79676	5913	383597.50	1645829.8	20.6565
	80	5	0.3259	42113	13725	176252.50	370847.3	8.8060
2010	0	1	0.0085	100000	854	99204.16	7537446.8	75.3745
	20	5	0.0130	98439	1281	488992.50	5558192.3	56.4633
	40	5	0.0152	93450	1422	463695.00	3640919.8	38.9612
	60	5	0.0614	83197	5105	403222.50	1852932.3	22.2716
	80	5	0.2878	49855	14347	213407.50	465542.3	9.3379
	0	1	0.0065	100000	653	99387.89	7711881.5	77.1188
	20	5	0.0078	98819	766	492180.00	5728579.0	57.9704
	40	5	0.0186	94320	1751	467222.50	3790881.5	40.1917

2020

60	5	0.0617	83360	5147	403932.50	1995551.5	23.9390
80	5	0.2433	51747	12590	227260.00	596549.0	11.5282

Table 12: Life Table Female Historic 1950-2021.

year	age	n	qx	lx	dx	Lx	Tx	ex
1950	0	1	0.0638	100000	6384	95054.13	6264224.5	62.6422
	20	5	0.0179	87316	1563	432672.50	4475656.4	51.2581
	40	5	0.0343	79398	2720	390190.00	2802671.4	35.2990
	60	5	0.0870	66034	5746	315805.00	1336126.4	20.2339
	80	5	0.3231	34073	11011	142837.50	301793.9	8.8573
2000	0	1	0.0091	100000	907	99156.09	8063351.6	80.6335
	20	5	0.0022	98643	218	492670.00	6085355.6	61.6907
	40	5	0.0086	97058	838	483195.00	4125083.1	42.5012
	60	5	0.0385	90968	3504	446080.00	2231488.1	24.5305
	80	5	0.2606	61814	16108	268800.00	638558.1	10.3303
2010	0	1	0.0066	100000	660	99381.47	8285586.1	82.8559
	20	5	0.0017	99152	168	495340.00	6300299.1	63.5418
	40	5	0.0070	97951	685	488042.50	4326456.6	44.1696
	60	5	0.0333	92636	3080	455480.00	2408504.1	25.9997
	80	5	0.2176	67700	14731	301672.50	744279.1	10.9938
	0	1	0.0077	100000	771	99279.76	8540751.4	85.4075
	20	5	0.0020	99038	199	494692.50	6557736.5	66.2143
	40	5	0.0065	97863	633	487732.50	4586749.0	46.8691
	60	5	0.0306	92929	2842	457540.00	2667989.0	28.7100

2020

80	5	0.1694	71131	12053	325522.50	976154.0	13.7233
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Table 13: Lifetable Male Projected Median 2022-2062.

year	age	n	qx	lx	dx	Lx	Tx	ex
2022	0	1	0.0076	100000	760	99289.82	7649798.5	76.4980
	20	5	0.0103	98649	1013	490712.50	5669175.1	57.4681
	40	5	0.0235	93291	2190	460980.00	3742990.1	40.1217
	60	5	0.0637	81449	5185	394282.50	1977570.1	24.2799
	80	5	0.2465	51233	12631	224587.50	607325.1	11.8542
2030	0	1	0.0054	100000	537	99494.94	7784331.0	77.8433
	20	5	0.0102	98986	1013	492397.50	5797768.2	58.5716
	40	5	0.0219	93783	2058	463770.00	3864000.7	41.2015
	60	5	0.0595	82534	4913	400387.50	2083288.2	25.2416
	80	5	0.2296	53698	12330	237665.00	678148.2	12.6289
2050	0	1	0.0022	100000	223	99788.37	8112778.8	81.1278
	20	5	0.0101	99441	1007	494687.50	6118475.7	61.5287
	40	5	0.0185	94629	1755	468757.50	4173360.7	44.1023
	60	5	0.0503	84789	4261	413292.50	2362935.7	27.8684
	80	5	0.1915	59417	11381	268632.50	880540.7	14.8197

Table 14: Lifetable Female Projected Median 2022-2062.

year	age	n	qx	lx	dx	Lx	Tx	ex
2022	0	1	0.0064	100000	635	99404.46	8580795	85.8080
	20	5	0.0017	99176	168	495460.00	6595226	66.5002
	40	5	0.0067	97918	657	487947.50	4621451	47.1972
	60	5	0.0312	92926	2896	457390.00	2702001	29.0769
	80	5	0.1742	71390	12438	325855.00	1009946	14.1469
2030	0	1	0.0046	100000	456	99570.13	8770231	87.7023
	20	5	0.0013	99418	130	496765.00	6780391	68.2008
	40	5	0.0057	98401	557	490612.50	4799759	48.7775
	60	5	0.0277	93972	2602	463355.00	2865791	30.4962
	80	5	0.1574	74355	11704	342515.00	1136484	15.2846
2050	0	1	0.0020	100000	199	99811.02	9243938	92.4394
	20	5	0.0007	99751	68	498585.00	7248238	72.6633
	40	5	0.0037	99152	366	494845.00	5257540	53.0251
	60	5	0.0206	95885	1976	474485.00	3298793	34.4036
	80	5	0.1217	80558	9803	378282.50	1495755	18.5674

Table 15: Lifetable Male Projected 97.5 Percentile 2022-2062.

year	age	n	qx	lx	dx	Lx	Tx	ex
2022	0	1	0.0056	100000	561	99472.73	7770137.5	77.7014
	20	5	0.0101	98953	1003	492257.50	5784189.6	58.4539
	40	5	0.0221	93759	2072	463615.00	3850869.6	41.0720
	60	5	0.0600	82445	4946	399860.00	2071214.6	25.1224
	80	5	0.2316	53432	12377	236217.50	669517.1	12.5303
2030	0	1	0.0021	100000	213	99797.80	8140909.2	81.4091
	20	5	0.0097	99463	968	494895.00	6146342.0	61.7953
	40	5	0.0182	94775	1727	469557.50	4199452.0	44.3097
	60	5	0.0497	85053	4226	414700.00	2384987.0	28.0412
	80	5	0.1892	59859	11328	270975.00	895652.0	14.9627
2050	0	1	0.0004	100000	43	99958.98	8809035.8	88.0904
	20	5	0.0090	99740	901	496447.50	6810481.3	68.2823
	40	5	0.0132	95819	1264	475935.00	4852648.8	50.6439
	60	5	0.0362	88344	3201	433717.50	2997296.3	33.9276
	80	5	0.1343	68957	9258	321640.00	1390408.8	20.1634

Table 16: Lifetable Female Projected 97.5 Percentile 2022-2062.

year	age	n	qx	lx	dx	Lx	Tx	ex
2022	0	1	0.0052	100000	520	99510.69	8696133	86.9613
	20	5	0.0015	99333	144	496305.00	6707795	67.5284
	40	5	0.0060	98228	594	489655.00	4729597	48.1492
	60	5	0.0290	93587	2714	461150.00	2800910	29.9284
	80	5	0.1638	73227	11997	336142.50	1085580	14.8249
2030	0	1	0.0025	100000	250	99762.93	9114028	91.1403
	20	5	0.0008	99687	81	498232.50	7119461	71.4181
	40	5	0.0041	98999	410	493970.00	5130756	51.8263
	60	5	0.0223	95457	2128	471965.00	3177346	33.2856
	80	5	0.1305	79047	10318	369440.00	1391666	17.6056
2050	0	1	0.0007	100000	67	99936.14	9926923	99.2692
	20	5	0.0003	99916	28	499510.00	7928347	79.3501
	40	5	0.0021	99621	207	497587.50	5932009	59.5458
	60	5	0.0139	97462	1351	483932.50	3954957	40.5795
	80	5	0.0860	86622	7451	414482.50	2084612	24.0656

Table 17: Lifetable Male Projected 2.5 Percentile 2022-2062.

year	age	n	qx	lx	dx	Lx	Tx	ex
2022	0	1	0.0103	100000	1032	99043.30	7523761.7	75.2376
	20	5	0.0104	98217	1018	488540.00	5550796.9	56.5156
	40	5	0.0249	92708	2310	457765.00	3634144.4	39.1999
	60	5	0.0675	80334	5425	388107.50	1885324.4	23.4686
	80	5	0.2623	48941	12836	212615.00	548861.9	11.2148
2030	0	1	0.0134	100000	1337	98771.73	7408036.4	74.0804
	20	5	0.0105	97703	1027	485947.50	5444271.3	55.7227
	40	5	0.0262	92066	2412	454300.00	3538698.8	38.4365
	60	5	0.0710	79245	5626	382160.00	1806891.3	22.8013
	80	5	0.2763	46900	12960	202100.00	501728.8	10.6978
2050	0	1	0.0117	100000	1173	98917.11	7467613.6	74.6761
	20	5	0.0108	97986	1055	487292.50	5498802.6	56.1182
	40	5	0.0255	92393	2357	456072.50	3587645.1	38.8303
	60	5	0.0692	79812	5521	385257.50	1847047.6	23.1425
	80	5	0.2691	47963	12908	207545.00	525575.1	10.9579

Table 18: Lifetable Female Projected 2.5 Percentile 2022-2062.

year	age	n	qx	lx	dx	Lx	Tx	ex
2022	0	1	0.0078	100000	776	99275.19	8463571.1	84.6357
	20	5	0.0020	98980	197	494407.50	6481475.0	65.4827
	40	5	0.0074	97549	726	485930.00	4513112.5	46.2651
	60	5	0.0335	92184	3086	453205.00	2604305.0	28.2512
	80	5	0.1852	69441	12860	315055.00	937522.5	13.5010
2030	0	1	0.0083	100000	832	99224.17	8421734.4	84.2173
	20	5	0.0021	98900	209	493977.50	6441061.7	65.1270
	40	5	0.0077	97403	752	485135.00	4474874.2	45.9419
	60	5	0.0343	91903	3153	451632.50	2570174.2	27.9662
	80	5	0.1892	68730	13005	311137.50	912751.7	13.2803
2050	0	1	0.0059	100000	591	99445.03	8621578.4	86.2158
	20	5	0.0016	99236	160	495780.00	6634958.7	66.8604
	40	5	0.0065	98033	634	488580.00	4659526.2	47.5302
	60	5	0.0304	93166	2831	458752.50	2736683.7	29.3743
	80	5	0.1705	72050	12284	329540.00	1036223.7	14.3820

Life Expectancy at Birth

Table 19: Life Expectancy at Birth Historic 1950-2021.

year	female	male	year	female	male	year	female	male
1950	62.6422	59.5195	1974	75.9569	69.2619	1998	79.6124	71.0135
1951	62.4997	59.5328	1975	77.0499	69.7210	1999	80.0908	72.0511
1952	64.7931	61.4149	1976	76.8520	69.6439	2000	80.6335	72.7017
1953	67.6486	64.4323	1977	77.0129	69.9331	2001	80.8430	72.4452
1954	69.4801	65.4902	1978	77.5263	70.0918	2002	81.5651	73.4655
1955	70.1169	66.4347	1979	77.4773	70.2506	2003	81.5182	73.8767
1956	70.0877	66.0901	1980	77.7493	70.5119	2004	81.5142	73.8632
1957	70.8638	66.8876	1981	77.8518	70.4326	2005	81.5023	73.7361
1958	71.1360	67.0596	1982	77.7453	70.9791	2006	82.4302	74.5202
1959	71.9985	67.0387	1983	78.4229	71.2995	2007	82.1284	74.4944
1960	72.0876	67.4948	1984	78.4778	71.6698	2008	82.4334	74.9099
1961	72.0318	67.3013	1985	78.3365	70.8567	2009	82.8543	74.7423
1962	72.4416	67.2679	1986	78.7237	70.7602	2010	82.8559	75.3745
1963	72.5762	66.5570	1987	78.7837	70.6493	2011	82.8879	75.1495
1964	71.8181	65.9274	1988	78.5353	70.1606	2012	83.2703	75.7190
1965	73.1115	67.0381	1989	78.3789	69.8420	2013	84.1614	76.3304
1966	73.9699	68.4819	1990	78.6651	69.9553	2014	83.9285	76.4006
1967	74.6482	68.8313	1991	79.1310	69.9185	2015	85.3228	77.8235
1968	74.7051	68.6002	1992	78.9586	69.5932	2016	84.9930	77.3234
1969	74.8432	68.5879	1993	78.6926	69.1756	2017	84.9643	76.6844
1970	74.9787	68.7273	1994	79.1232	69.6016	2018	86.3209	77.9377
1971	75.4525	69.2082	1995	78.6311	68.9282	2019	86.2360	77.9945

1972	75.3145	68.6690	1996	78.9239	69.8960	2020	85.4075	77.1188
1973	75.8018	68.8184	1997	79.5694	70.9987	2021	85.5671	76.3280

Table 20: Life Expectancy at Birth Projected Median 2022-2062.

year	female	male	year	female	male	year	female	male
2022	85.8080	76.4980	2036	89.1090	78.8320	2050	92.4394	81.1278
2023	86.0459	76.6707	2037	89.3422	78.9943	2051	92.6855	81.2921
2024	86.2851	76.8385	2038	89.5771	79.1567	2052	92.9291	81.4595
2025	86.5226	77.0069	2039	89.8190	79.3140	2053	93.1737	81.6217
2026	86.7605	77.1760	2040	90.0542	79.4818	2054	93.4209	81.7856
2027	86.9955	77.3461	2041	90.2865	79.6416	2055	93.6637	81.9608
2028	87.2331	77.5162	2042	90.5254	79.8090	2056	93.9100	82.1221
2029	87.4677	77.6759	2043	90.7580	79.9707	2057	94.1582	82.2888
2030	87.7023	77.8433	2044	90.9976	80.1327	2058	94.4088	82.4596
2031	87.9376	78.0116	2045	91.2385	80.2989	2059	94.6584	82.6247
2032	88.1694	78.1792	2046	91.4792	80.4653	2060	94.9075	82.7934
2033	88.4056	78.3370	2047	91.7228	80.6315	2061	95.1654	82.9564
2034	88.6409	78.5043	2048	91.9579	80.7982	NA	NA	NA
2035	88.8761	78.6671	2049	92.1993	80.9627	NA	NA	NA

Table 21: Life Expectancy at Birth Projected 97.5 Percentile 2022-2062.

year	female	male	year	female	male	year	female	male
2022	86.9613	77.7014	2036	93.6412	83.5022	2050	99.2692	88.0904
2023	87.6706	78.3694	2037	94.0430	83.8397	2051	99.6727	88.4141
2024	88.2702	78.9080	2038	94.4464	84.1702	2052	100.0929	88.7360
2025	88.8134	79.3809	2039	94.8435	84.5013	2053	100.5045	89.0609
2026	89.3350	79.8408	2040	95.2537	84.8390	2054	100.9022	89.4065
2027	89.8153	80.2443	2041	95.6449	85.1586	2055	101.3227	89.7396
2028	90.2666	80.6414	2042	96.0540	85.4775	2056	101.7474	90.0760
2029	90.7050	81.0373	2043	96.4426	85.8017	2057	102.1703	90.4130
2030	91.1403	81.4091	2044	96.8456	86.1239	2058	102.6193	90.7389
2031	91.5811	81.7706	2045	97.2514	86.4444	2059	103.0736	91.0864
2032	91.9962	82.1301	2046	97.6423	86.7711	2060	103.4971	91.4355
2033	92.4204	82.4789	2047	98.0477	87.0899	2061	103.9168	91.7983
2034	92.8313	82.8196	2048	98.4485	87.4267	NA	NA	NA
2035	93.2279	83.1606	2049	98.8588	87.7497	NA	NA	NA

Table 22: Life Expectancy at Birth Projected 2.5 Percentile 2022-2062.

year	female	male	year	female	male	year	female	male
2022	84.6357	75.2376	2036	84.6269	74.0554	2050	86.2158	74.6761
2023	84.3822	74.8802	2037	84.7159	74.0758	2051	86.3667	74.7468
2024	84.2500	74.6453	2038	84.8227	74.0868	2052	86.4885	74.8294
2025	84.1797	74.4759	2039	84.9060	74.1296	2053	86.6478	74.8885
2026	84.1358	74.3428	2040	85.0110	74.1509	2054	86.7749	74.9688
2027	84.1370	74.2457	2041	85.1264	74.2017	2055	86.9040	75.0356
2028	84.1528	74.1669	2042	85.2402	74.2405	2056	87.0447	75.1254
2029	84.1801	74.1133	2043	85.3615	74.2990	2057	87.1769	75.2081
2030	84.2173	74.0804	2044	85.4799	74.3427	2058	87.3188	75.2744
2031	84.2610	74.0327	2045	85.6048	74.4113	2059	87.4595	75.3678
2032	84.3305	74.0367	2046	85.7205	74.4587	2060	87.6147	75.4233
2033	84.3884	74.0189	2047	85.8334	74.5059	2061	87.7541	75.5082
2034	84.4645	74.0368	2048	85.9698	74.5608	NA	NA	NA
2035	84.5407	74.0396	2049	86.0968	74.6217	NA	NA	NA

Fitted parameters A_x and B_x

Table 23: Fitted Ax and Bx Parameters Puerto Rico 1950-2021

Age	Ax Male	Bx Male	Ax Female	Bx Female
0	-3.9748	0.1333	-4.1503	0.0793
1	-7.6576	0.2594	-7.6248	0.1600
5	-8.2748	0.1746	-8.4284	0.1110

10	-7.9462	0.0974	-8.4576	0.0727
15	-6.7076	0.0228	-7.8245	0.0658
20	-6.0681	0.0026	-7.3742	0.0633
25	-5.9114	0.0089	-7.0787	0.0601
30	-5.7647	0.0166	-6.7920	0.0509
35	-5.5627	0.0200	-6.4619	0.0449
40	-5.3381	0.0259	-6.1856	0.0406
45	-5.0889	0.0232	-5.8534	0.0322
50	-4.7448	0.0244	-5.4527	0.0308
55	-4.4143	0.0206	-5.0851	0.0261
60	-4.0558	0.0261	-4.6401	0.0284
65	-3.6846	0.0276	-4.1915	0.0295
70	-3.2795	0.0331	-3.6956	0.0321
75	-2.9360	0.0264	-3.2867	0.0251
80	-2.4490	0.0304	-2.7068	0.0262
85	-1.8365	0.0268	-1.9743	0.0208

Fitted parameter K_t

Table 24: Fitted Kt Parameter Puerto Rico 1950-2021

year	male	female	year	male	female	year	male	female
1950	11.4705	19.5594	1974	2.7150	4.1242	1998	-0.6528	-2.7517
1951	11.5477	19.9037	1975	2.2815	2.0211	1999	-1.7761	-3.7138
1952	11.0294	18.4223	1976	2.4356	2.2949	2000	-2.8561	-4.8768
1953	10.0871	16.6144	1977	2.1461	2.1214	2001	-2.8216	-5.4874
1954	9.2420	15.0778	1978	1.7281	1.2599	2002	-4.3060	-7.1282
1955	8.5011	14.2564	1979	1.7893	1.2274	2003	-4.8205	-7.0904
1956	8.3838	13.9104	1980	1.4189	0.7070	2004	-4.7894	-7.0605
1957	7.6820	12.5424	1981	1.5217	0.4512	2005	-4.7729	-6.9936
1958	7.8766	12.4055	1982	0.6880	0.6561	2006	-6.5170	-9.1194
1959	7.4054	11.5600	1983	0.4747	-0.5148	2007	-6.1982	-8.6270
1960	6.8690	11.1422	1984	-0.2395	-0.6205	2008	-7.0378	-9.3654
1961	6.6529	10.7496	1985	0.5114	-0.4391	2009	-7.0139	-10.4101
1962	6.8238	10.2978	1986	0.0928	-0.9892	2010	-8.1326	-10.6367
1963	7.0063	10.3371	1987	0.1443	-1.0649	2011	-8.0388	-10.7559
1964	7.8979	11.5723	1988	0.3116	-0.6808	2012	-8.9933	-11.6642
1965	6.6761	9.7454	1989	0.8019	-0.2044	2013	-10.4728	-13.6932
1966	5.4195	7.9778	1990	0.6143	-0.7049	2014	-9.9747	-13.3295
1967	4.7798	6.9872	1991	0.2798	-1.5647	2015	-13.2141	-16.5945
1968	4.3967	5.9298	1992	0.5449	-1.1503	2016	-12.4802	-15.9399
1969	4.0813	5.9790	1993	0.7571	-0.5728	2017	-11.6876	-15.9235
1970	4.1823	5.6947	1994	-0.0067	-1.6664	2018	-14.2803	-19.3285

1971	3.6541	5.0363	1995	1.0274	-0.6590	2019	-14.8125	-19.5937
1972	3.9232	5.2970	1996	0.0625	-1.2103	2020	-12.3906	-17.5685
1973	3.2400	4.1335	1997	-0.7929	-2.5597	2021	-12.0962	-17.7409

Code and Source Data

All code and data used in this analysis is available in the GitHub repository